

Cosmic Ray Shower Development By Element at 10^{18} eV

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Abstract

Cosmic rays are particles from space, made of a mix of protons and atomic nuclei. Everyday, single ultra-high-energy primary atoms ($>10^{18}$ eV) induce particle showers to rain down on Earth, and the question of the primaries' exact composition remains a focus in particle astrophysics. When the particles interact they create generations of further particles. As the energy is depleted throughout the shower, the production slows. This inflection point, known as $X_{\max}$, correlates to the number of protons in the primary cosmic particle. The goal of this research is to simulate showers from different nuclear species using CORSIKA, in order to ascertain descriptive statistics of their $X_{\max}$, and to make these descriptions available to the community.

Introduction

Ultra-high energy cosmic rays, originating from outer space, shower Earth's atmosphere constantly at nearly the speed of light. These cosmic rays are mostly comprised of protons or heavier nuclei [1].



Fig 1. Artists interpretation of a cosmic ray shower over the Pierre Auger Observatory in Argentina. The added blue arrow indicates the direction of shower depth.

After the initial, primary cosmic ray begins to shower, it will produce generations of further particles, drawing from its initial energy. The creation of these secondary particles will hit a peak in the atmosphere called $X_{\max}$ which corresponds to the atomic number of the primary cosmic ray [2]. Elements heavier than iron are unlikely to survive the cosmic acceleration process, fragmenting into smaller elements.

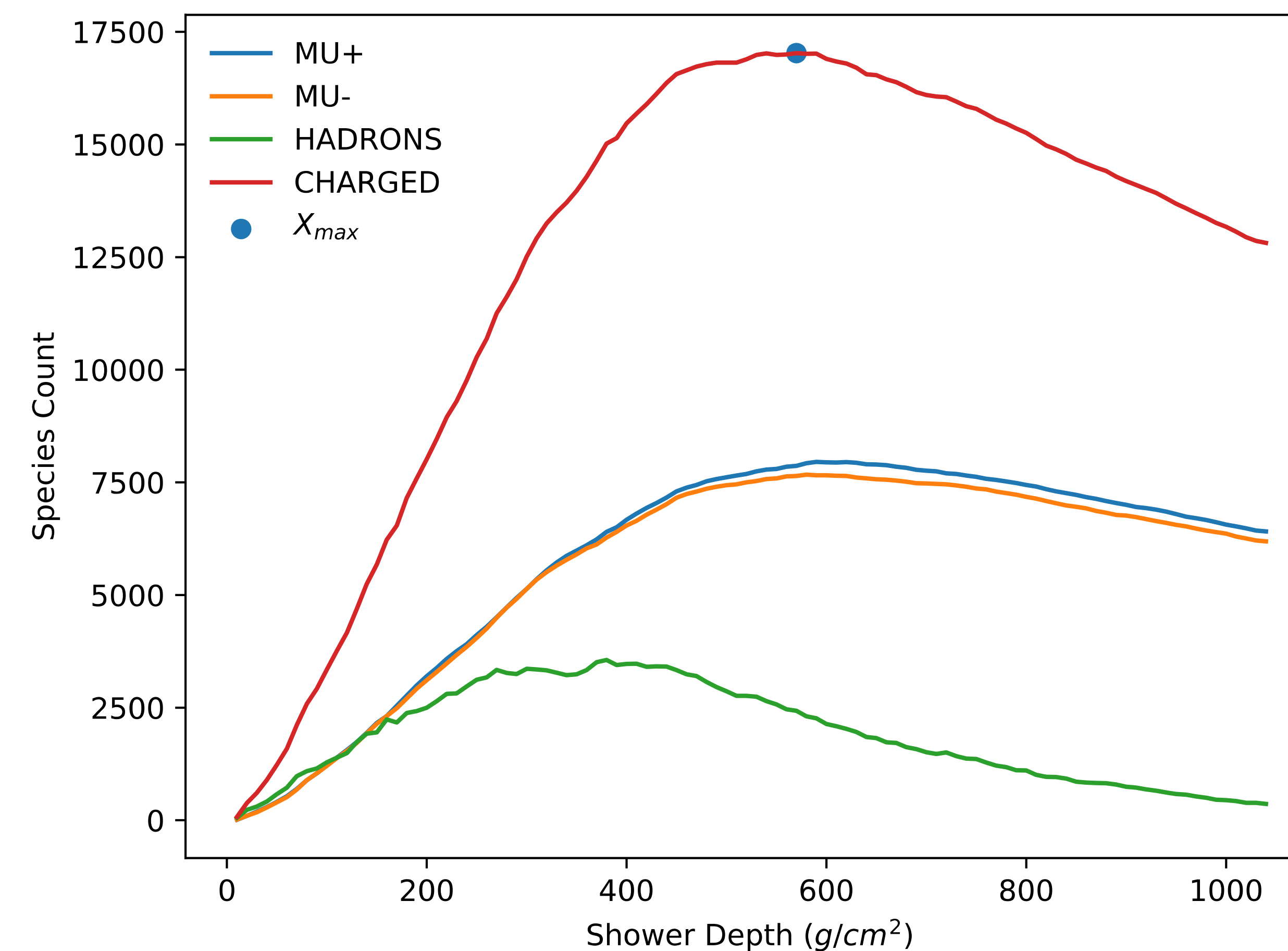


Fig 2. Longitudinal development of a cosmic ray shower for a regular Iron nucleus at 10^{15} eV. This plot quantifies the number of particles of each of the types produced in the shower as a function of the depth in the shower. Depth here is measured in grammage, which starts at zero at the top of the atmosphere and integrates the density of matter crossed by the shower.

Methods

CORSIKA, a computer software for the simulation of extensive air showers induced by high-energy cosmic rays, was compiled on an Ubuntu-based homemade cluster [3]. Steering files representing the twenty six lightest elements (choosing the most abundant isotope) were made, and events were run for each element and saved. To extract and aggregate data to produce the graphs, python-based methods were employed.

Each shower was run with an initial primary energy of 10^{18} eV, and initial direction aimed directly down through the atmosphere. Atmospheric and ambient magnetic conditions are taken from central Europe.

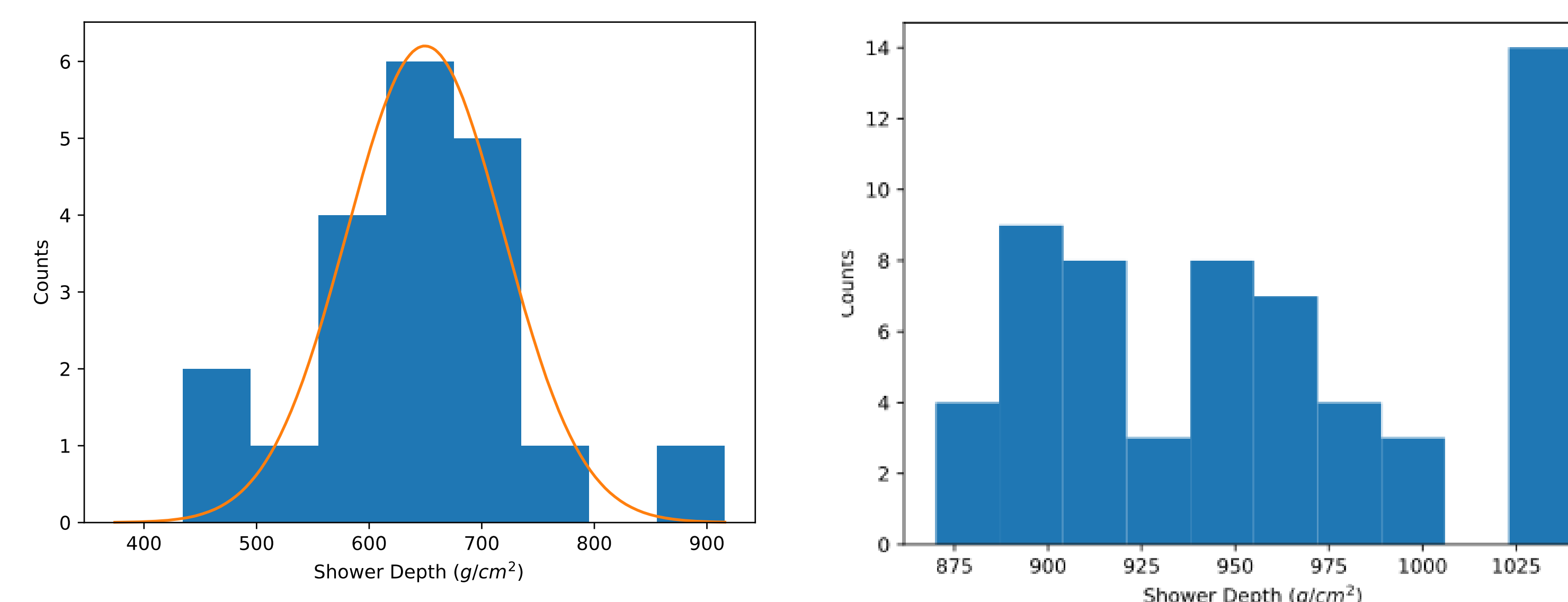


Fig 3. Left: a histogram of the $X_{\max}$ values obtained from our simulation for Lithium at 10^{15} eV, showing a fairly shallow $X_{\max}$. The means and standard deviations of each of these elemental histograms represent a single datapoint in Figure 4. Right: a similar histogram for Lithium at 10^{18} eV.

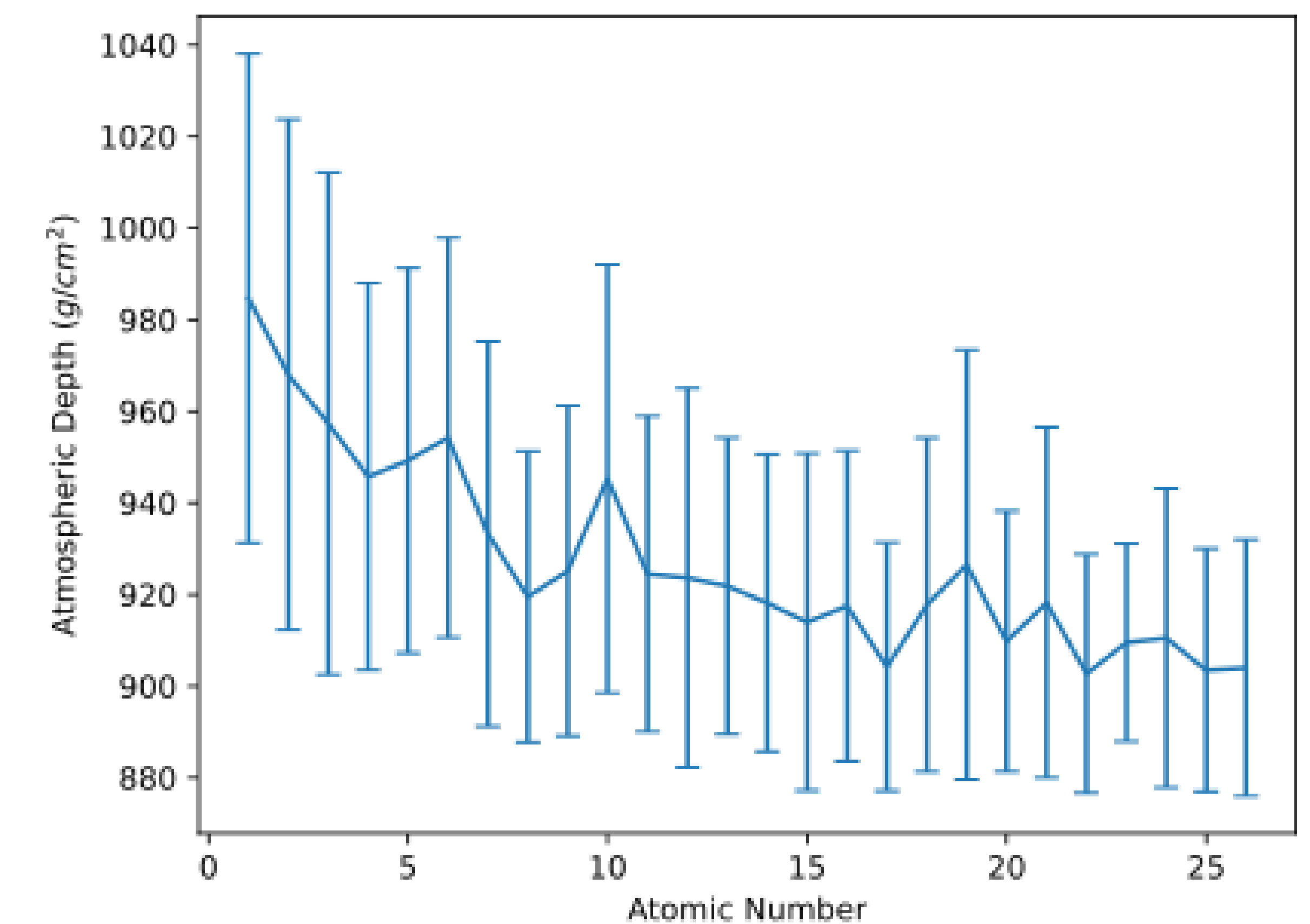


Fig 4. A final plot of the $X_{\max}$ of the first 26 elements. The trend towards having interactions deeper in the atmosphere as the elements get heavier is expected, as more nucleons create a higher chance of interaction at lower depths.

Results and Conclusions

While the trend we see in our data has been known in the particle astrophysics community for some time, a clear visualization of $X_{\max}$ as a function of atomic number has not been widely available. Figure 4 shows a clear trend towards lesser depths for higher atomic number, confirming the intuition that having more nucleons to start the initial shower causes the shower to begin higher in the atmosphere. Now that we have explored higher energies than the previous work, we will begin simulating a library of various energies and atomic number.

References

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- [3] D, H., & J, K. (February 1998). Kit - corsika - corsika 7 - documentation. <https://www.iap.kit.edu/corsika/70.php>

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